

Practical Work 03 – 03/03/2026

Model Selection

Objectives

The main objectives of this Practical Work for Week 3 are the following :

- a) Use an MLP model for classifying Fashion MNIST images. Build the model in pytorch with one hidden layer with 200 units. Verify that you can transition from the overfitting regime (starting training with just a single batch of 64 samples) to a regime with better generalisation by increasing the number batches used for training.
- b) Evaluate MLPs with different number of layers and units per layer for FashionMNIST image classification. Identify an MLP for which you obtain optimal validation performance (accuracy), on the full training dataset, without using regularisation.
- c) Evaluate whether you can gain performance (accuracy) by tuning the complexity and regularisation (dropout, L2).
- d) Optional : Play through an example of overfitting and determine the optimal model complexity.

Submission

- **Deadline** : Tuesday 10 March, 15 :00
- **Format** : Jupyter notebooks including comments and results. For the solution of Exercises 1-3, you will use one single notebook.
- Please **only one submission per group** - with a clear indication of the group id and the names of the participants.

Exercise 1 Evaluation of an MLP for Fashion MNIST with Increasing Training Set

In this (and in the two subsequent) exercises, you will build an MLP for Fashion MNIST classification. As a first step, you will train an MLP with a single hidden layer with only one single batch (batchsize 32). This is a good test indicating whether you have properly implemented the model and the training procedure. Then, gradually increase the training set and see whether you will arrive in a situation where there is no longer overfitting. Specify how you identify overfitting. Use the notebook `mlps_for_fashion_MNIST_stud.ipynb` (Section "Fixed MLP with Increasing Training Dataset") that provides the skeleton for the exercise and some additional details.

Exercise 2 Evaluation of an MLP for Fashion MNIST with Different Model Complexities

Continue from the previous exercise - but now playing with different model complexities while using the full training data. Again use the notebook `mlps_for_fashion_MNIST_stud.ipynb` (Section "Evaluate Different Model Complexities") that provides the skeleton for the exercise and some additional details.

Exercise 3 Evaluation of an MLP for Fashion MNIST with Different Model Complexities and with Regularisation

Continue from the previous exercise by using a model where you can observe overfitting. Now add regularisation. Try to tune the complexity of the model and regularisation parameters to obtain a best MLP model. Again use the notebook `mlps_for_fashion_MNIST_stud.ipynb` (Section "Add Regularisation") that provides the skeleton for the exercise and some additional details.

Exercise 4 Optional : Bias-Variance Tradeoff

Evaluate the bias and variance error for the polynomial models as presented in the lecture. Follow the instruction in the notebook `overfitting_polynomial_stud.ipynb`

Exercise 5 Optional : Review Questions

- a) Explain the terms bias error and variance error. What are the factors that make the bias error larger or how can it be made smaller? What factors lead to a large variance error or how can it be reduced?

- b) Why is the training error an increasing function of the split ratio (fraction of samples used for training)? Why is the validation error a decreasing function of the split ratio?
- c) What is understood under the phenomenon of "deep double decent" ?
- d) What is the goal of regularisation in deep learning?
- e) List four regularisation method and describe in two sentences each the underlying idea and how they work.